CleanSweep Ultra AI User Manual



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Introduction

Thank you for purchasing the CleanSweep Ultra AI by SmartHome Robotics. This advanced robotic vacuum and mop system represents the cutting edge of home cleaning technology. Powered by artificial intelligence and equipped with state-of-the-art sensors, the CleanSweep Ultra AI navigates your home with precision, adapting to different floor types and avoiding obstacles with remarkable accuracy. This user manual will guide you through the setup, operation, and maintenance of your new cleaning companion to ensure you get the most out of its capabilities.

Getting Started

Package Contents

- CleanSweep Ultra AI robot vacuum
- Self-emptying base station
- Power cord
- Side brush (2)
- HEPA filter (1 pre-installed, 1 spare)
- Mopping pad (1 pre-installed, 1 spare)
- Water tank (pre-installed)
- Dust collection bags (3)
- Cleaning tool

- Quick start guide
- This user manual

Before First Use

1. Remove all packaging materials and protective films
2. Install side brushes if not already attached
3. Place the self-emptying base station in an appropriate location
4. Charge the robot fully before initial use (approximately 3 hours)
5. Download the SmartHome app from the App Store or Google Play Store
6. Remove any small objects, cords, or other potential obstacles from floors

Product Overview

 CleanSweep Ultra AI Components

Robot

1. LiDAR sensor dome
2. Control buttons
3. Bumper with collision sensors
4. RGB camera
5. 3D ToF sensors
6. Dust bin release button
7. Charging contacts
8. Cliff sensors (bottom)
9. Main brush
10. Side brushes
11. Drive wheels
12. Omnidirectional wheel
13. Water tank
14. Mopping pad

Base Station

1. Dust collection port
2. Bag compartment
3. Status indicators
4. Charging contacts
5. Power connection
6. Cord storage

Technical Specifications

Feature	Specification
Suction Power	4500Pa with auto carpet boost
Battery	5200mAh lithium-ion, up to 180 minutes runtime

Feature	Specification
Dust Bin	400ml onboard, 3L capacity in base station
Water Tank	300ml electronic water control system
Mapping Technology	LiDAR + RGB camera + 3D ToF sensors
Navigation	AI-powered SLAM with real-time obstacle avoidance
Connectivity	Wi-Fi, Bluetooth 5.0, voice assistant compatible
Noise Level	58-68 dB (adjustable)
Climbing Ability	Up to 2cm threshold crossing
Dimensions	Robot: 13.8" diameter x 3.8" height, Base: 12" x 16" x 19"
Weight	Robot: 8.2 lbs, Base station: 11.5 lbs

Safety Instructions

Important Safeguards

- Read all instructions before using the robot vacuum
- Use only as described in this manual
- Do not use outdoors or on wet surfaces
- Do not handle the charging dock or robot with wet hands
- Do not use without dust bin and filters in place
- Keep hair, loose clothing, fingers, and all parts of body away from openings and moving parts
- Do not use to pick up flammable or combustible liquids
- Do not use to pick up anything that is burning or smoking
- Do not use without the dust bin and/or filters in place
- Turn off all controls before unplugging
- Do not place objects on top of the robot
- Be extra careful when cleaning stairs
- Do not use to pick up hard or sharp objects such as glass, nails, screws, or coins
- Use extra care when cleaning drapes, tassels, or delicate items
- Keep the robot away from heated surfaces

Child and Pet Safety

- Close supervision is necessary when used around children or pets
- Do not allow to be used as a toy
- The robot's sensors will detect and avoid most obstacles, but small pets should be supervised during operation
- Consider using no-go zones or scheduling cleaning when pets are not in the area

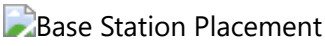
Setup and Installation

Base Station Placement

1. Place the base station on a hard, level surface against a wall
2. Ensure there is at least 1.5 feet (0.5 meters) of clearance on both sides and 4 feet (1.2 meters) in front of the base
3. Place the base station within range of a Wi-Fi signal
4. Avoid placing near stairs or drop-offs
5. Connect the power cord to the base station and plug into a wall outlet

Robot Preparation

1. Turn the robot over and attach the side brushes by pressing them onto the mounting posts until they click
2. Turn the robot right-side up
3. Place the robot on the base station to charge
4. The robot will beep and the power indicator will pulse while charging
5. A full charge takes approximately 3 hours

 Base Station Placement

Mobile App Setup

Downloading the App

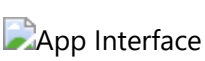
1. Search for "SmartHome Robotics" in the App Store (iOS) or Google Play Store (Android)
2. Download and install the app
3. Create an account or log in if you already have one

Connecting Your Robot

1. Ensure your smartphone is connected to your home Wi-Fi network (2.4GHz networks only)
2. Open the SmartHome app
3. Tap the "+" icon to add a new device
4. Select "CleanSweep Ultra AI" from the list
5. Press and hold the Power and Home buttons on the robot simultaneously for 3 seconds until you hear a beep
6. Follow the on-screen instructions to connect the robot to your Wi-Fi network
7. Once connected, you can name your robot and access all features through the app

Initial Mapping

1. After setup, tap "Create Map" in the app
2. The robot will begin a mapping run to create a detailed floor plan of your home
3. For best results, ensure all doors are open to rooms you want mapped
4. The robot will return to the base station when mapping is complete
5. You can then customize the map by naming rooms and creating virtual boundaries

 App Interface

Basic Operation

Control Buttons

- **Power**: Press to start/pause cleaning; press and hold for 3 seconds to power on/off
- **Home**: Press to send the robot back to the base station
- **Spot Clean**: Press to start intensive cleaning of a specific area

Starting a Cleaning Cycle

1. Press the Power button on the robot or tap "Clean" in the app
2. The robot will leave the base station and begin cleaning
3. By default, the robot will clean the entire mapped area and return to the base when finished

Pausing and Stopping

1. Press the Power button on the robot or tap "Pause" in the app to pause cleaning
2. To resume, press the Power button again or tap "Resume" in the app
3. To end the cleaning cycle, press the Home button or tap "Dock" in the app

Manual Control

1. In the app, tap "Manual Control"
2. Use the directional pad to steer the robot
3. Tap "Exit" to return to automatic cleaning mode

Cleaning Modes

Auto Clean

The default mode that intelligently cleans your entire home based on the created map.

Room Clean

1. In the app, tap "Room Clean"
2. Select the specific rooms you want to clean
3. Tap "Start" to begin cleaning only those rooms

Zone Clean

1. In the app, tap "Zone Clean"
2. Draw a box around the specific area you want to clean
3. Tap "Start" to begin cleaning only that zone

Spot Clean

1. Place the robot near the soiled area
2. Press the Spot Clean button on the robot or tap "Spot Clean" in the app
3. The robot will clean in a spiral pattern, intensively cleaning a 3ft x 3ft area

Scheduled Cleaning

1. In the app, tap "Schedule"
2. Tap "Add Schedule"
3. Select the days, time, rooms, and cleaning mode

4. Toggle the schedule on/off as needed

Mapping and Navigation

Multi-Floor Mapping

The CleanSweep Ultra AI can store up to 4 different floor plans.

1. In the app, tap "Maps" > "Add New Map"
2. Take the robot to the new floor and place it on the floor (not on the base station)
3. Tap "Start Mapping"
4. The robot will create a new map of the floor
5. The robot will automatically recognize which map to use based on its location

No-Go Zones

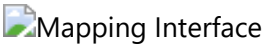
1. In the app, tap "Maps" > select the map > "Edit"
2. Tap "No-Go Zone" and draw rectangles around areas you want the robot to avoid
3. Tap "Save" to apply changes

Virtual Walls

1. In the app, tap "Maps" > select the map > "Edit"
2. Tap "Virtual Wall" and draw lines to create boundaries the robot won't cross
3. Tap "Save" to apply changes

Furniture and Obstacle Detection

- The robot uses 3D sensors to detect and navigate around furniture and obstacles
- The RGB camera helps identify and avoid common household items like shoes, pet bowls, and cords
- For best results, pick up small items and secure loose cords before cleaning

 Mapping Interface

Mopping System

Preparing for Mopping

1. Remove the water tank from the back of the robot
2. Open the water filling port and fill with clean water only (no cleaning solutions)
3. Close the port securely
4. If not already attached, snap the mopping pad onto the bottom of the water tank
5. Slide the water tank back into the robot until it clicks

Mopping Settings

1. In the app, tap "Cleaning Preferences" > "Mopping"
2. Select the water flow level (Low, Medium, or High)
3. Enable or disable "Auto-Avoid Carpets" feature
4. For specific rooms, you can customize the water flow level in the room settings

After Mopping

1. Remove the water tank
2. Empty any remaining water
3. Remove and rinse the mopping pad
4. Allow the mopping pad to air dry completely before reattaching
5. Clean the water tank monthly by rinsing with warm water

Mopping Tips

- For best results, vacuum the floor before mopping
- Use only water in the tank; cleaning solutions may damage the system
- Remove the mopping attachment when not in use
- Replace mopping pads every 2-3 months depending on usage

Self-Emptying Base Station

Auto-Empty Function

- After each cleaning cycle, the robot will return to the base station
- The base station will automatically begin extracting dust and debris from the robot's dust bin
- This process takes approximately 15-20 seconds and may be louder than normal operation
- The dust is collected in a sealed bag inside the base station

Dust Bag Replacement

1. Lift the top cover of the base station
2. Pull up on the dust bag handle to remove the full bag
3. The bag will automatically seal when removed
4. Insert a new dust bag, ensuring it clicks into place
5. Close the top cover

Base Station Maintenance

- Check the dust bag monthly and replace when full
- Clean the charging contacts on both the robot and base station monthly with a dry cloth
- Check the dust extraction port for clogs if you notice reduced suction


Dust Bag Replacement

Voice Assistant Integration

Amazon Alexa Setup

1. In the Amazon Alexa app, tap "Skills & Games"
2. Search for "SmartHome Robotics"
3. Tap "Enable Skill"
4. Log in with your SmartHome account credentials
5. Discover devices by saying "Alexa, discover devices"

Google Assistant Setup

1. Open the Google Home app
2. Tap "+" > "Set up device" > "Works with Google"
3. Search for "SmartHome Robotics"
4. Log in with your SmartHome account credentials
5. Your robot will appear in the Google Home app

Voice Commands

- "Start cleaning" - Begins a cleaning cycle
- "Stop cleaning" - Pauses the current cleaning cycle
- "Return to dock" - Sends the robot back to the base station
- "Clean the [room name]" - Cleans a specific room
- "What is the battery level?" - Reports the current battery status

Maintenance

Regular Maintenance Schedule

Component	Maintenance Action	Frequency
Dust bin	Empty (if not using auto-empty base)	After each use
Filters	Tap to remove dust	Weekly
Filters	Replace	Every 2 months
Side brushes	Clean	Monthly
Side brushes	Replace	Every 3-6 months
Main brush	Clean	Monthly
Main brush	Replace	Every 6-12 months
Sensors	Clean	Monthly
Charging contacts	Clean	Monthly
Front wheel	Clean	Monthly
Dust bag (base station)	Replace	Every 60 days (when full)
Mopping pad	Clean	After each use
Mopping pad	Replace	Every 2-3 months

Cleaning the Dust Bin (Manual)

1. Press the dust bin release button and pull to remove
2. Open the dust bin lid and empty contents into a trash bin
3. Rinse the dust bin with water if necessary
4. Allow to dry completely before reinstalling

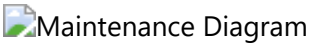
5. Note: If using the self-emptying base, the dust bin will be emptied automatically

Cleaning the Main Brush

1. Turn the robot over
2. Press the brush guard tabs and lift to remove the cover
3. Lift out the main brush
4. Clean hair and debris from the brush and brush compartment
5. Reinstall the brush and brush guard

Cleaning the Sensors

1. Turn the robot over
2. Wipe the cliff sensors on the bottom with a clean, dry cloth
3. Turn the robot right-side up
4. Wipe the LiDAR sensor dome with a clean, dry cloth
5. Clean the camera lens and 3D sensors on the front with a clean, dry cloth


Maintenance Diagram

Troubleshooting

Problem	Possible Cause	Solution
Robot won't turn on	Battery depleted	Place on charging dock and ensure charging contacts are clean
Robot won't charge	Poor connection to base	Clean charging contacts on robot and base station
Poor cleaning performance	Brushes tangled or dustbin full	Clean brushes and empty dustbin
Robot gets stuck frequently	Cluttered environment	Remove small objects from floor and use no-go zones
Robot can't find base station	Base station poorly positioned or blocked	Ensure base station is against wall with clear path
App can't connect to robot	Wi-Fi issues	Ensure robot is within range of Wi-Fi and restart router
Mopping leaves streaks	Dirty mopping pad or improper water flow	Replace mopping pad or adjust water flow settings
Base station not emptying robot	Clogged extraction port or full dust bag	Check for clogs or replace dust bag
Error message: "Wheels suspended"	Robot is lifted or stuck	Place robot on flat surface and remove any obstacles
Error message: "Brush blocked"	Main brush is tangled	Clean the main brush and remove any debris

Error Codes

Code	Meaning	Solution
E001	Wheel suspended	Place robot on flat surface
E002	Bumper stuck	Tap bumper to dislodge any debris
E003	Cliff sensor blocked	Clean cliff sensors
E004	Low battery	Place robot on charging dock
E005	Brush blocked	Clean main brush
E006	Side brush blocked	Clean side brushes
E007	Lidar sensor blocked	Check and clean LiDAR sensor dome
E008	Dustbin missing	Install dustbin
E009	Dustbin full	Empty dustbin
E010	Network error	Check Wi-Fi connection

Factory Reset

If you experience persistent issues, perform a factory reset:

1. Ensure the robot is powered on
2. Press and hold the Power and Home buttons simultaneously for 10 seconds
3. The robot will beep and all lights will flash
4. The robot will restart and you will need to set it up again

Warranty Information

The CleanSweep Ultra AI comes with a 2-year limited warranty covering defects in materials and workmanship under normal household use. For warranty service, contact SmartHome Robotics customer support with your proof of purchase.

What's Covered

- Robot vacuum and components
- Base station and components
- Battery (1 year)

What's Not Covered

- Normal wear and tear (brushes, filters, etc.)
- Damage from misuse or abuse
- Damage from improper maintenance
- Commercial use

Contact Support

- Customer Service: 1-800-SMARTHOME (1-800-762-7846)
- Email: support@smarthomerobotics.com
- Website: www.smarthomerobotics.com/support
- Live Chat: Available 24/7 through the SmartHome app

For tutorials, tips, and additional information, visit our website or access the Help section in the SmartHome app.

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